

Glossary

AC (alternating current)

AC is an electrical current that repeatedly changes in direction.

acceleration

An increase or decrease in an object's velocity (speed) due to a force being applied to it.

acid

A compound that breaks up into a negative ion and one or more positive hydrogen ions, which react easily with other substances.

activation energy

The energy needed to start a chemical reaction.

air resistance

A force that pushes against an object that is moving through the air, slowing it down; also called drag.

algae

Plantlike organisms that live in water or damp habitats; in general, they are single-celled.

alkali

A compound that dissociates into negative hydroxide (OH) ions and a positive ion; alkalis react easily with acids.

allotrope

A variant form of an element; for example, carbon can occur as graphite or diamond; while allotropes look different and have various physical properties, they all have identical chemical properties.

alloy

A mixture of two or more metals, or a metal and a nonmetal.

amplitude

The height of a wave.

anatomy

The science that studies the structure of living bodies to discover how they work.

anion

A negatively charged ion formed when an atom or group of atoms gains one or more electrons.

arthropod

A member of the largest animal phylum, which includes spiders, insects, and crustaceans.

atmosphere

A blanket of gases that surrounds a planet, moon, or star.

atom

The smallest unit of an element.

atomic number

The number of protons located in the nucleus of an atom; every element has atoms with a unique atomic number.

attraction

A force that pulls things together; opposite of repulsion.

bacteria (singular: bacterium)

Single-celled organisms that form a distinct kingdom of life; compared to other cells, bacterial cells are small and lack organelles.

base

An ionic compound that reacts with an acid.

biomass

A way of measuring the total mass of living things in a certain region; a useful way of comparing different types of organism in an ecosystem.

boiling point

The temperature at which a heated substance changes from

a liquid into a gas; when the gas is cooled, it will condense into a liquid at this same temperature.

buoyancy

The tendency of a solid to float or sink in liquids.

capillary

A small blood vessel that delivers oxygen to body cells.

catalyst

A substance that lowers the activation energy of a chemical reaction, making the reaction occur much more rapidly.

cations

Positively charged ions, which form from atoms (or molecules) that lose one or more electrons.

cell

The smallest unit of a living body.

cellulose

A complex carbohydrate that makes up the wall that surrounds all plant cells.

chemical

A pure substance that has distinct properties.

chlorophyll

The green-colored compound that collects the energy in sunlight so it can be used to react with carbon dioxide and water to make sugar during photosynthesis.

chromosome

A structure in the nucleus of cells that is used to store coils of DNA.

circuit

A series of components (such as light bulbs) connected between the poles of a battery or other power source so an electric current runs through them.

combustion

A chemical process in which a substance reacts with oxygen, releasing heat and flames.

compound

A chemical that is made up of the atoms of two or more elements bonded together.

compression

Squeezing or pushing a substance into a smaller space.

concave

Having a curved surface that resembles the inside of a circle or sphere.

concentration

The amount of one substance mixed into a known volume of another.

condense

To turn from a gas to a liquid; for example, steam condenses into water.

conduction

The process by which energy is transferred through a substance. The energy being transferred is thermal (heat), acoustic, or electrical.

convection

A process that transfers heat through a liquid or gas, with warm areas rising and cooler ones sinking, thus creating a circulating current.

convex

Having a curved surface that resembles the outside of a circle or sphere.

current

A flow of a substance; electrical currents are a flow of electrons or other charged particles.